

LING 510: Week 3 review

La La (Lambda) land

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Office hours: W 11–1PM, N421H | [Zoom](#)



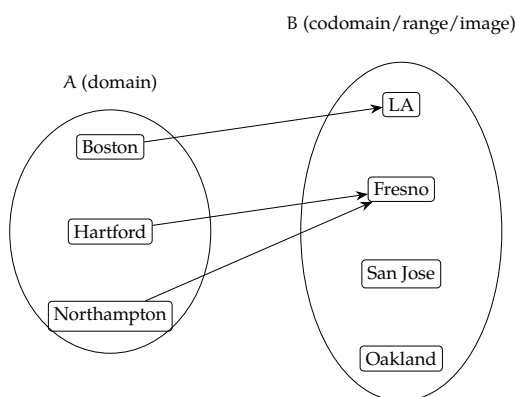
Plan for the day: Assignment 1, Tools for semantics: [for docx](#)

1 Mathematical preliminaries

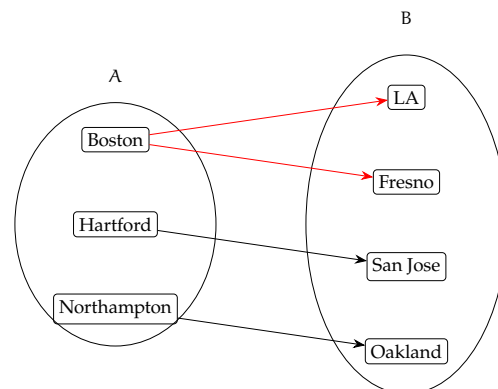
The tacit assumption for us is that even though language and its rules might seem arbitrary, it has a logical structure underlyingly. Our job is to uncover and describe this structure as accurately as possible. We have made good on the first step – 1. Compositionality 2. Syntax 3. Meaning as truth conditions. Now we will develop our formal machinery and have a toolkit to get to task.....by the power of Math. Turns out, most everything when it comes to meaning can be modeled in terms of functions from set of things to set of things.¹ So we need to know what functions are.

1.1 Functions

A function is like a map. It tells you what location you will land on if you start from a given location. For the map to be 'good', it should be unambiguous i.e., it should tell you exactly where you will land given a starting location. In other words, **each starting location must have a unique (i.e., one and only one) landing location**. This is shown below, where **1a** describes a 'well-defined' function and **1b**, which is not a function.



(a) This **is** a function: Boston, Hartford, Northampton each send out exactly one arrow. (It's fine that San Jose, Oakland receive nothing, and that Hartford, Northampton both point to Fresno.)



(b) This is **not** a function: Boston sends out **two** arrows (to LA and Fresno), so it's ambiguous where you would "land" from Boston.

If I wanted to send the map to someone else, how would I do it? I can think of a couple of

¹Functions are also sets of things so we can do this recursively. Its super-effective!

ways of doing this: 1. Listing the pairs of starting and landing sits 2. A table 3. Something more compact. They all represent the same phenomenon.

$$1. f = \{ \langle \text{Boston}, \text{LA} \rangle, \langle \text{Hartford}, \text{Fresno} \rangle, \langle \text{Northampton}, \text{Fresno} \rangle \}$$

$$2. f = \begin{bmatrix} \text{Boston} \rightarrow \text{LA} \\ \text{Hartford} \rightarrow \text{Fresno} \\ \text{Northampton} \rightarrow \text{Fresno} \end{bmatrix}$$

$$3. \text{ For } A = \{\text{Boston}, \text{Hartford}, \text{Northampton}\}, B = \{\text{LA}, \text{Fresno}, \text{San Jose}, \text{Oakland}\}, \\ f : A \rightarrow B \text{ such that } f(\text{Boston}) = \text{LA}, f(\text{Hartford}) = f(\text{Northampton}) = \text{Fresno} \quad (\text{better!})$$

Let's do better. I can actually describe the domains and codomains in a way that covers what is common to all the members in respective *sets* i.e., I can describe them by way of a property.

$$(1) \quad \text{Let } A = \{x : x \text{ is a town in Massachusetts}\}, B = \{y : y \text{ is a town in California}\}, \\ f : A \rightarrow B \text{ such that } f(\text{Hartford}) = f(\text{Northampton}) = \text{Fresno}$$

Takeaway: A function from a set A to set B is a **set** of *ordered pairs* where the first member (from A) in each ordered pair has a unique corresponding member from set B. A function is a well-defined relation on the set $A \times B$ (the set of all ordered pairs from A to B). These two are different ways of talking about functions. They are logically equivalent.

Comprehension check:

1. Is 'cookie monster $\in \{x : x \text{ is blue}\}$ ' true?
2. What are the members of the set $\{x : \text{the sky is blue}\}$?
3. $\{x : x \text{ is blue}\} \cap \{x : x \text{ has four wheels}\} = ?$
4. Is $\{x : x \text{ is blue}\} \subseteq \{x : x \text{ is long}\}$?
5. $\{x : x \text{ is blue}\} \cap \{\{\text{cookie monster}\}\} = ?$
6. Does Cookie monster $\in \{y : \{y\} \cap \{x : \{x\} \subseteq \{x : x \text{ is a muppet}\} \cap \{x : x \text{ is blue}\}\} \neq \emptyset$?
7. Is it true that for $f: \{y : y \in \mathbb{N} \text{ and } y \leq 5\} \rightarrow \mathbb{N}$, for every $x \in \{y : y \in \mathbb{N} \text{ and } y \leq 5\}$, we have that $f(x) = 2x$
8. $f : \mathbb{N} \rightarrow \mathbb{N}$ we have that $f(x) = \sqrt{(x)}$, is this a function?
9. Write the following function $\{\langle 4, 2 \rangle, \langle 6, 3 \rangle, \langle 10, 5 \rangle, \langle 14, 7 \rangle, \langle 22, 11 \rangle, \langle 26, 13 \rangle, \langle 34, 17 \rangle\}$ in the table format and set abstraction format.

1.2 Bring on the lambdas

We can write our functions by listing all its members – 1. as set of ordered pairs 2. a table describing the pair-wise mappings of members from domain to range. The more economical way is to just define the function by a criterion that captures all mappings. One of them is (most likely) familiar to you all. The other is **lambda notation**. We will be using the latter because, as you will see through the course, it is extremely powerful in helping us express very complex meanings and build them piece-by-piece.

1. $F := f : \mathbb{N} \rightarrow \mathbb{N}$ for every $x \in \mathbb{N}$, $f(x) = x^2$
2. $F := \{ \langle x, y \rangle : y = x^2 \}$ (for $x, y \in \mathbb{N}$)
3. $F := [\lambda x : x \in \mathbb{N}. x^2]$

All of the above describe an input condition² that must be met for the function to be defined or computable, the value description that captures relation between input value and output value, and the variables that introduces/instantiate those values.

$$\begin{array}{ccc} \text{introduces} & \longrightarrow & \lambda x : \underbrace{x \in \mathbb{N}}_{\text{input condition}} \cdot x^2 \longleftarrow \text{value description} \\ \text{the variable} & & \end{array}$$

Here's how you can compute values in lambda:

- (2) a. $F(7) = f(7) = (7)^2 = 49$
- b. $F(7) = [\lambda x : x \in \mathbb{N}. x^2](7)$
 $= (7)^2$
 $= 49$

$([\lambda x : x \in \mathbb{N}. x^2](7))$
 $\uparrow \beta\text{-reduction}$

Function application in lambda notation is done by a step called β -reduction where, if the input condition on the variable is met, you erase the input condition and all instances of the variable with the input value. Otherwise, it fails, and value is undefined.

- (3) Let $F := [\lambda y : y \in \mathbb{N}. \sqrt{y}]$ then
 $[\lambda y : y \in \mathbb{N}. \sqrt{y}](\text{Gazpacho})$ is **undefined**.

You can have complex lambda expressions that introduce multiple variables such as,

- (4) $F := [\lambda x : x \in \mathbb{N}. \lambda y : y \in \mathbb{Z}. x^y]$

To see why order matters, consider the following minimal pair,

- (5) a. $(F(-3))(2)$
 $= ([\lambda x : x \in \mathbb{N}. \lambda y : y \in \mathbb{Z}. x^y](-3))(2)$
 $= [\lambda y : y \in \mathbb{Z}. 2^y](-3)$
 $= 2^{-3}$
 $= 1/2^3 = 1/8$
 $(1/8 \in \mathbb{Q}, \text{ the set of rational numbers})$
- b. $(F(2))(-1)$
 $= ([\lambda x : x \in \mathbb{N}. \lambda y : y \in \mathbb{Z}. x^y](2))(-1)$

undefined, since $-1 \notin \mathbb{N}$

Notice that when you first β -reduce, you still have a lambda expression with a variable! Which means that the resulting output from the first β -reduction is a function. The function in (4) is **a function from natural numbers to a function from integers (to rational numbers)**. Lambda expressions are important for us because, as we will see, natural language expressions way too often denote complex functions like (4).

²The input condition is *sort of* like how you'd imagine a presupposition to be!

Practice with lambdas:

1. Write the function $\{ \langle 1, 1 \rangle, \langle 2, 4 \rangle, \langle 3, 9 \rangle, \langle 4, 16 \rangle \}$ (for $x \in \mathbb{N}, x \leq 4$) in lambda notation.
2. Write $F := [\lambda x : x \in \{y : y \in \mathbb{N}, y \leq 5\}. 2x + 1]$ (for $x \leq 5$) in table format and in set of ordered pairs format. Compute $F(2)$ and $F(23)$.
3. Let $F := [\lambda x : x \in \mathbb{Z}. 126]$. What is $F(-370)$?
4. Let $F := [\lambda x : x \in \mathbb{N}. x^2]$. Compute $F(\text{Elmo})$. Is the result defined? Why or why not?
5. Let $F := [\lambda x : x \in \mathbb{N}. \lambda y : y \in \mathbb{N}. x \times y]$. Compute $(F(3))(5)$, showing every β -reduction step. What kind of function is left over right after the *first* reduction? Describe the complex function F in words.
6. Let $F := [\lambda x : x \in \mathbb{Z}. \lambda y : y \in \mathbb{N}. x - y]$. Compute $(F(-2))(4)$ and $(F(4))(-2)$. Is either undefined? What does this tell you about why the order of the introduced variables matters?
7. **Challenge:** Write a lambda expression F that takes a natural number x , then a natural number y , and returns *True* just in case x is greater than y , and *False* otherwise. Then compute $(F(7))(2)$ and $(F(2))(7)$. Can you write in words what function this expression represents? Can you write it in the table format?

<i>Teletubby</i>	Situation <i>s</i>	
	<i>Occupation</i>	<i>Preference</i>
Tinky-winky	leader	red bags, playtime
Dipsy	student	dancing, playtime
Laa-laa	dancer, student	singing and dancing, playtime
Po	student	driving, playtime
Mi-Mi	baby	sleeping, playtime
Da-Da	baby	sleeping
Baa	baby	sleeping and singing
Ru-Ru	baby	singing
Ping	baby	sleeping

Table 1: teletubbies

8. Define the following function using lambda notation:

$$f : \{x : x \text{ is a Teletubby}\} \rightarrow \{x : x \text{ is a set of preferences}\}$$

for every $z \in \{x : x \text{ is a Teletubby}\}$, $f(z)$ is z 's set of preferences.

9. Consider the table above (representing a situation s), and give the following values:

- (a) $[\lambda x : x \in \{y : y \text{ is a baby in } s\}. \text{True if } x\text{'s preference is playtime; else, False}](\text{Mi-Mi})$
- (b) $[\lambda x : x \in \{y : y \text{ is a Teletubby in } s\}. \text{the number of preferences } x \text{ has}](\text{Laa-laa}) +$
 $[\lambda x : x \in \{y : y \text{ is a Teletubby in } s\}. \text{the number of preferences } x \text{ has}](\text{Mi-Mi})$
- (c) $[\lambda x : x \in \{y : y \text{ is a preference in } s\}. \text{False if } x \text{ starts with the letter } s; \text{True otherwise}]([\lambda x : x \in \{y : y \text{ is a Teletubby in } s\}. x\text{'s preference}](\text{Da-Da}))$

10. Consider the table above, and define the following function as a set of ordered pairs:

$[\lambda x : x \in \{y : y \text{ is a Teletubby in } s\}. \text{the color yellow if } x \text{ is a baby; else, the color blue}]$